

Ryan Qin

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Education

University of Waterloo

Sept 2024 - Jun 2029

Bachelor of Applied Science in **Electrical Engineering**, Honours, Co-op

Technical Skills

Languages: C, C++, C#, Assembly, Verilog, Java, Python, HTML/CSS, JavaScript

Technologies: Robotics, RTOS, Linux, STM32, DSP, FSM, FPGA, TCP/IP, MongoDB, Node.js, React

Development Tools: TensorFlow, PyTorch, Android, AWS, Git, Cadence, Altium, MATLAB, DSO, Fusion 360

Professional Experiences

I/O Compliance Engineer | TensorFlow, OpenCL, C++, C#, Cadence, DSO, I²C

May 2026 - Present

AMD

- Developing a **TensorFlow** model for predicting root cause of **DP** and **HDMI** test failure, aiming to reduce debugging time by **50%**
- Applying **autoencoder ML** to HDMI protocol analyzer for **improving** effectiveness of **detecting** frame glitches and stutters compared to **standard** test patterns and manual monitor observations
- Designing **STM32-based I²C** module for **automatic** HDMI **EDID** setting, helping to reduce manual rewiring work
- Optimizing electrical Compliance analysis calculations using **OpenCL**, utilizing GPU parallelism for consecutive test runs

Electrical Engineer | C++, OpenCV, TCP/IP, LtSpice

Sep 2025 - Dec 2025

ATS Corporation

- Managed **full-cycle** design and verification for **four parallel** electronics projects, ensuring **on-time, spec-compliant** delivery
- Implemented **OpenCV** jet spray analysis, reducing test time by **60%** through automated spray angle and stability measurements
- Developed **C++** firmware for a custom **STM32-based** leak-testing control board with **customized** hardware drivers, improving power consumption by **40%** while decreasing unit's size by **80%**
- Designed and implemented **PID control loops** and **dither circuits** for a fuel nozzle testing system's pressure regulator, enabling **25%** more accurate simulation of turbojet engine operating conditions.

Electrical Coordinator | C#, .Net, MySQL, COMSOL, Python

Jan 2025 - May 2025

WSP

- Recognized by **WSP Canada's** Rail and Transit VP for OCS engineering Software's **innovation intake** potential
- Independently developed Overhead Contact System (**OCS**) engineering software in **C#**, reducing Calculation time by **40%**
- Optimized traction power substation's cooling system layout in **COMSOL FEA**, aimed to extend transformer life by 10 years
- Implemented AI-driven weather prediction using **Prophet**, improving storm and temperature tolerance by **30%**

Projects

FPGA Clock 📁 | Verilog, Vivado, FSM, FPGA

Dec 2025 - Jan 2026

- Developed a **real-time** VGA digital clock for FPGA using **Verilog**, designing **Moore FSM** for time counters, VGA timing, and character rendering using Vivado and HDL simulation
- Implemented parameterized **RTL** to simplify character resizing and repositioning across display resolutions

Autodump 📁 | I/O system Design, C++, TCP/IP, SSH, Python, OpenCV

Nov 2025

- Developed an **autonomous** self-emptying trash robot, achieving an **overall 3rd** place at **MakeCU**
- Designed **ArUco** marker navigation system in **OpenCV**, allowing the robot to **autonomously navigate** back to home position
- Engineered an Arduino-based serial **distributed I/O** system, focused on **simplifying** software-hardware integration and **reducing PWM-signal jitter** and **integrity** for servo and ultrasonic sensors

Competition RC 📁 | PyTorch, FreeRTOS, LTSpice, AVR Assembly

May 2025

- Built an RC car with **50%** recycled material; selected as the **top qualified finalist** out of 15 teams
- Implemented a **lane-keeping algorithm**, combining **PID** and **PyTorch**-based vision to adjust steering from USB camera
- Designed a **custom** airsoft motor controller using MOSFET stages, relays, and Arduino for PWM signal conversion, achieving a **fast lap record** of **17 seconds** compared to **average 25 seconds** lap time

Baymax Bot 📁 | Machine Learning, Python, C++, STM32, TCP/IP

Jan 2025

- Trained a voice model using **Applio**, replicating BayMax's Voice with **customized response** from **ChatGPT's API**
- Implemented a UART-based **STM32 field I/O** controller to **offload** time-critical motor coordination from a **Raspberry Pi**